

Parametric Equations

ANSWER KEY



Eliminating the parameter

- 1 Consider the curve $x = 2t + 1$, $y = t^2$.
 - a Eliminate the parameter to write y as a function of x . $t = \frac{x-1}{2}$, so $y = \frac{(x-1)^2}{4}$.
 - b Identify the curve. A parabola with vertex $(1, 0)$ opening upward.
 - 2 Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \leq t < 2\pi$, and describe the curve.
 $x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$: a circle of radius 3 centered at the origin, traced counterclockwise.
 - 3 For the line $x = t - 2$, $y = 2t + 3$:
 - a Find the points corresponding to $t = 0$ and $t = 2$. $t = 0$: $(-2, 3)$; $t = 2$: $(0, 7)$.
 - b Eliminate the parameter to write the Cartesian equation. $t = x + 2$, so $y = 2x + 7$.
-

Projectile motion

- 4 A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).
 - a When does the projectile land ($y = 0$, $t > 0$)? $-5t(t - 6) = 0$, so $t = 6$ s.
 - b Find the horizontal range. $x(6) = 240$ m.
 - c Find the maximum height. Peak at $t = 3$: $y(3) = -45 + 90 = 45$ m.